

GALAXIES

A HUNDRED billion galaxies exist in the Universe. Each consists of a vast collection of stars, gas, and dust. They started life thousands of million of years ago, slowly forming into distinctive shapes. Each galaxy can contain billions of stars. Gravity keeps the stars together and keeps the galaxies in clusters.

Types of galaxies

Most galaxies have a central ball of stars, the nucleus, and many have a flattened disc coming out of this. Astronomers have classified galaxies into three main types based on these features. No one knows why galaxies become a particular shape. It may be to do with how fast a galaxy spins and how quickly stars form inside.

Elliptical

About 60 per cent of galaxies are ball-shaped collections of old stars. They range in shape from round to flattened ovals. Astronomers describe their shape with the letter *E* followed by a number between 0 and 7 – the higher the number, the flatter the galaxy.

Irregular

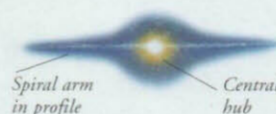
About 10 per cent of galaxies are irregular. They are collections of stars with no distinctive shape or structure, and do not fit into any of the classifications. They are smaller than the average galaxy and contain large amounts of gas and dust.

Barred spiral

These galaxies consist of a central bar of older stars with arms containing younger stars coming from the ends of the bar. Barred spirals are described as *SB* followed by a letter from *a* to *d* to indicate how tightly wound the arms are and the size of the hub.

Spiral

A hub of older stars is surrounded by a flattened disc with spiral arms containing younger stars. The shape of a spiral is described by the letter *S*, followed by a letter between *a* and *d* to indicate how tightly wound the arms are and the size of the hub.



The Milky Way is about 100,000 light years wide and 13,000 light years across at the central hub.

Active galaxies

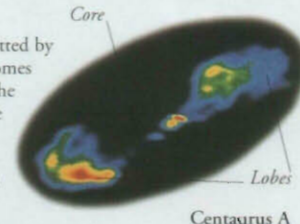
Unusually large amounts of energy are emitted from some galaxies. This energy may come from an object that is visible, such as a quasar, or from an invisible object, such as the lobes of a radio galaxy. How the energy is created is uncertain, but evidence suggests it could be from a supermassive black hole at the centre of the galaxy.

Quasars

Quasars are the brightest, most distant, fastest moving, and youngest objects visible outside the Milky Way. Thousands are known, each emitting huge amounts of energy. They are found at the heart of large galaxies.

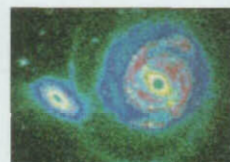
Radio galaxies

Powerful radio energy is emitted by radio galaxies. The energy comes from lobes at either side of the visible core, and is detectable with radio telescopes. Centaurus A is the nearest active galaxy: it is 16 million light years away.



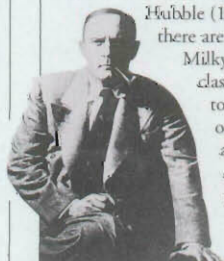
Colliding galaxies

Galaxies can collide as they move through space, as is happening (left) with two galaxies in the constellation of Boötes. Such collisions will change the shape of a galaxy or result in a merger.



Edwin Hubble

In 1923, the American astronomer Edwin Hubble (1889–1953) proved that there are galaxies other than the Milky Way. The next year he classified galaxies according to their shape. He went on to show that galaxies are moving away from each other, and so provided proof that the Universe is expanding.



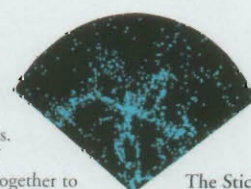
Galaxy clusters

Galaxies are grouped together in clusters. The Milky Way belongs to a cluster of about 30 galaxies called the Local Group. The Virgo Cluster (right) contains about 2,500 galaxies, mostly spirals.



Superclusters

Clusters of galaxies group together into superclusters that spread across many millions of light years. In turn, hundreds of superclusters group together to form huge walls and filaments, such as the Stick Man, which stretch for hundreds of millions of light years.



The Stick Man contains millions of galaxies.

FIND OUT
MORE

ASTRONOMY

BIG BANG

BLACK HOLES

GRAVITY

STARS

UNIVERSE